Version of Record: https://www.sciencedirect.com/science/article/pii/S0378112723002487
Manuscript_5400bcbc6fe975412547530974c69587

Mapping territorial vulnerability to wildfires: a participative multi-criteria analysis

Miguel Rivière[*,1,2], Jonathan Lenglet[3], Adrien Noirault[3], François Pimont[4], Jean-Luc Dupuy[4]

[*] corresponding author : miguel.riviere@agroparistech.fr, 22 place de l'agronomie 91120 Palaiseau France

[1] Université Paris-Saclay, AgroParisTech, CNRS, Ecole des Ponts ParisTech, CIRAD, EHESS, UMR CIRED, 94130, Nogent-sur-Marne, France.

[2] Université de Lorraine, Université de Strasbourg, AgroParisTech, CNRS, INRAE, BETA, Nancy 54000 France

[3] Université de Lorraine, AgroParisTech, INRAE, UMR Silva, Nancy, France

4 Ecologie des Forêts Méditerranéennes (URFM), INRAE, F-84914 Avignon, France

Highlights

- Analytical Hierarchy Process proves a flexible method to assess vulnerability and risk
- Risk experience, wildland-urban interface and tree traits drive vulnerability
- Considering socio-economic drivers and several assets broadens the analysis
- Stakeholders confirmed the relevance of participative assessments using MCDA
- We identify vulnerable locations not yet subject to hazard

Abstract

The Mediterranean region is routinely affected by forest fires, with adverse consequences on ecological, infrastructural, and socioeconomic assets. In a context of climate change, it is

© 2023 published by Elsevier. This manuscript is made available under the Elsevier user license
https://www.elsevier.com/open-access/userlicense/1.0/

crucial for fire prevention and suppression to be able to identify locations where assets are most at risk, due to environmental, physical or socio-economic reasons. Besides, this knowledge needs to be developed conjunctly with operational services and field experts to ensure their application. So far, fire risk research has largely focused on fire behavior, effects, and model simulation. In this article, we show how the concept of vulnerability can provide a flexible and relevant framework for assessing fire risk and be evaluated using a spatial multicriteria decision analysis method (MCDA) - the Analytical Hierarchy Process (AHP)- based on both quantitative data and expert judgment gathered through a participative approach. We focus on South-Eastern France, a region characterized by high economic and environmental stakes and heavily affected by wildfires. We develop a series of spatialized indicators using ecological, land-use and sociodemographic data which we aggregate to produce vulnerability maps for three categories of assets: population, ecosystems and infrastructures. An ex-post workshop was organized with field experts to put both the approach and results into discussion. Results reveal significant differences in indicators' perceived contributions to risk and vulnerability and enable appraising the contribution of sociodemographic factors, often overlooked in the literature. We also reveal differences in spatial patterns across both vulnerability subcomponents and exposed assets, helping identify primary and secondary vulnerability hotspots and underlying drivers. Consideration of multiple subcomponents of risk and vulnerability may help local decision makers prioritize how and where measures should be implemented, while the use of MCDA favors experience and knowledge sharing and among stakeholders while providing a basis of discussion.

Keywords

AHP, GIS-MCDA, wildfire, vulnerability, risk assessment, forest

1. Introduction

Human societies derive benefits from well-functioning ecosystems through the services they provide (Diaz et al., 2019). Hazards such as storms, fires and pest outbreaks are natural ecosystem features whose regimes (i.e. patterns of occurrence and intensity) are expected to be affected by global changes (IPCC, 2021; Seidl et al., 2017). Wildfires, or forest fires, are one of the commonest disturbances to affect vegetated ecosystems (Bowman et al., 2020) and can negatively affect environmental, socio-economic, and infrastructural assets (Gill et al., 2013; J. C. Liu et al., 2015). Their occurrence and behavior have environmental determinants such as weather and fuel moisture, as well as anthropogenic determinants such as land-use patterns and the presence of nearby activities (Bowman et al., 2020; Ganteaume et al., 2013). Most of these are dependent on the local context, e.g. wildfires are known to be prevalent in locations with high wildland-urban interface (Radeloff et al., 2018). As a result, there is a need to discuss and prepare fire risk management policy locally with stakeholders, especially in regions where wildfire regimes are expected to worsen, such as the Mediterranean (Dupuy et al., 2020; Pimont et al., 2022).

As a field of study, wildfire risk assessment is concerned with the integrative evaluation of fire likelihood and impacts (Oliveira et al., 2021). Wildfire risk is often defined as the resultant of three characteristics: *fire occurrence* and *behavior* - their resultant being *fire likelihood*, i.e. the probability that a fire will happen - and *fire effects* on values-at-risk (Finney, 2005; Thompson et al., 2013). The evaluation of fire risk and its components have been made possible by the development of sophisticated wildfire simulators alongside fire effects models (Costafreda-Aumedes et al., 2017; O'Brien et al., 2018; Xi et al., 2019). Despite the existence of this "technical" definition of wildfire risk, a large part of the literature either uses unclear definitions or puts a strong emphasis on fire hazard, i.e. the

physical phenomenon, through metrics such as burn probability and fire intensity (Johnston et al., 2020), ignoring the impact on values-at-risk.

The natural hazards literature, especially that related to climate change (e.g. Füssel & Klein, 2006; Metzger et al., 2006) adopts a broader definition of risk as "*the potential for consequences where something of value is at stake and where the outcome is uncertain*" (IPCC, 2019). Within this framework, vulnerability refers to "*the propensity or predisposition to be adversely affected by a hazard, including sensitivity or susceptibility to harm and lack of capacity to cope and adapt*". Applications of this framework, despite shifting contours, share a common point in the consideration of four key aspects: hazard, exposure of assets, their susceptibility (or sensitivity) and adaptive capacity (also, coping capacity or resilience) (Comte et al., 2019). Despite their similarities, this latter conceptualization is more encompassing than the traditional view of wildfire risk due to the clear distinction made between exposure of assets and hazard likelihood, the explicit inclusion of adaptive capacity and the consideration of social and economic dynamics. Besides, clearer boundaries are set to vulnerability, while the term can be found in the fire risk literature with a meaning akin to sensitivity (Johnston et al., 2020; Liu et al., 2012). While some recent studies have adopted this broader view (Ghorbanzadeh et al., 2019; Lecina-Diaz, Martínez-Vilalta, Alvarez, Vayreda, et al., 2021; Oliveira et al., 2018, 2020; Thorne et al., 2010), the general framework has only recently been formally adapted to the case of forest-related natural hazards (Lecina-Diaz, Martínez-Vilalta, Alvarez, Banqué, et al., 2021; Oliveira et al., 2018; Vaillant et al., 2016).

Many fire risk assessments use model simulations, raising questions of model integration, the use of a common (often monetary) metric, and, given model complexity, the difficulty to transfer knowledge to local decision makers. Model-based assessments also tend to overlook

or simplify human and social dimensions, even though recent advances have been made regarding e.g. citizen involvement and behavioral aspects (Johnston et al., 2020; Oliveira et al., 2021). This can be addressed with multiple-criteria decision analysis (MCDA), which relates to an ensemble of methods that propose a structured and formal framework for assessing complex problems considering several dimensions. MCDA methods can integrate data of mixed natures, including expert knowledge, and foster collaborative decision making by providing focus to discussions. MCDA is particularly well-suited for natural resources management issues, especially at local scales (Mendoza & Martins, 2006). Their further integration into Geographic Information Systems (GIS) enables the combination of spatial data and value judgements to create visual information supporting decision making. GIS-MCDA methodologies have proven to be powerful, comprehensive and convenient tools in environmental planning when spatial drivers are significant (Malczewski, 2007; Malczewski & Rinner, 2015), including applications to hazard risk assessment for e.g. floods, earthquakes and explosions (Duzgun et al., 2011; Fernandez et al., 2015; Ghajari et al., 2017; Le Cozannet et al., 2013; Nyimbili et al., 2018; Pham et al., 2021). Owing to the emphasis on the physical phenomenon and model simulation, similar examples are scarcer in the fire risk literature. Ghorbanzadeh et al. (2019) use GIS-MCDA to assess social and infrastructural vulnerability based on census, land-use and economic data. They subsequently demonstrate the coupling of vulnerability maps to hazard maps generated with machine learning models to assess overall fire risk. Oliveira et al. (2020) focus on risk to populations and jointly assess hazard exposure and vulnerability using a fire spread simulator and a clustering algorithm on sociodemographic and geographic data. They also show how their model could be used to inform policy planning by evaluating overall risk under various scenarios of land management policy. Another example is given by Thompson et al. (2013), which consider several exposed assets and expert knowledge in their assessment of vulnerability.

In this article, our objective is to perform an encompassing assessment of fire risk based on the vulnerability concept, using field knowledge from experts with direct experience with wildfires and considering three assets of different natures important at the territorial scale: population, ecosystems and infrastructures. A secondary objective is to explore and confirm the potential of a MCDA participative approach to foster discussion between experts that do not usually engage in dialogue, in a field where model-based evaluations by academics constitutes the main research methodology. As a result, our approach is exploratory and is not a predictive assessment: the research process is as important as results, which are themselves (e.g. maps) tools to foster discussion and the emergence of collaborative knowledge.

Vulnerability is decomposed into a hierarchical problem and its subcomponents – exposure, sensitivity and adaptive capacity – are assessed by integrating quantitative land-use, economic and sociodemographic data as well as expert knowledge and simulation results using the Analytic Hierarchy Process (Saaty, 2004), a MCDA method based on stakeholder participation and expert judgment. This approach of vulnerability is furthermore combined with probabilistic modelling of fire activity (Pimont et al. 2021) to assess fire hazard, hence resulting in an integrated multidisciplinary approach to wildfire risk. We focus on South-Eastern France, a fire prone area with contrasted land-use patterns which influence fire activities (Castel-Clavera et al., 2022) and where fire regimes are expected to worsen through significant intensification of fire activities, an expansion of the fire-prone region and a lengthening of the summer fire season (Pimont et al. 2022).

We aim to develop a methodology suited to our biophysical and institutional context and to provide synthetic, visual and interpretable results at the meso scale. The availability of such resources can have implications for local stakeholders engaged in fire risk management, helping them identify vulnerability hotspots and underlying drivers, and to select measures to

be implemented locally. These latter aspects were discussed in an ex-post workshop with field experts.

2. Material and methods

2.1. General framework

2.1.1. Presentation of the study area

We focus on South-Eastern France (Fig. 1), namely the area known as *Prométhée* after the fire observation database covering it (https://www.promethee.com/). The region is mostly Mediterranean but displays varied topographic conditions with coastal plains, the Rhône river valley and several mountainous areas (e.g. the Alps[1]). It also includes the mountainous and highly forested Corsica island. Forests cover 30% of land area, agriculture 49% and artificial land uses 5% (Figure 1). Fires in the area are very numerous and an average of 9000 ha have burned yearly over in the period 2011-2021, representing 80% of the national total. Most fires occur over the summer (May-October) and are largely human-caused, voluntarily or accidentally. They tend to occur close to population centers, in areas with high wildland-urban or wildland-agriculture interfaces, and in locations with touristic frequentation (Ganteaume et al., 2013; Ruffault & Mouillot, 2017). Over the decades to come, fire danger is expected to increase significantly owing to increases in temperatures and changes in precipitation patterns (Dupuy et al., 2020; Fargeon et al., 2020). Climate change could induce an increase in the number of large fires (>100ha) by up to +180% for a +4°C warming, a doubling of fire-prone areas and a large symmetrical lengthening of the summer fire season (Pimont et al. 2022). Stakes in terms of exposed assets are high. There are over 9 million inhabitants in the area (14% of France's population), most of which are located in the lowlands, which also includes large economic and industrial centers. The region comprises 5 national parks and more than 15 regional parks with valuable habitats, as well as numerous

[1] An annotated map of the study area is provided in Appendix B.

reserves. Owing to the mountainous nature of many locations, in addition to classical ecosystem services (e.g. wood production, carbon storage), forests also provide important regulation services (e.g. erosion and flood control), as well as aesthetic and cultural services due to touristic frequentation, which is very high in the summer along the coast and in the mountains. These Land-Use and Land-Cover factors affect spatial patterns of fire activity (Castel-Clavera et al., 2022).

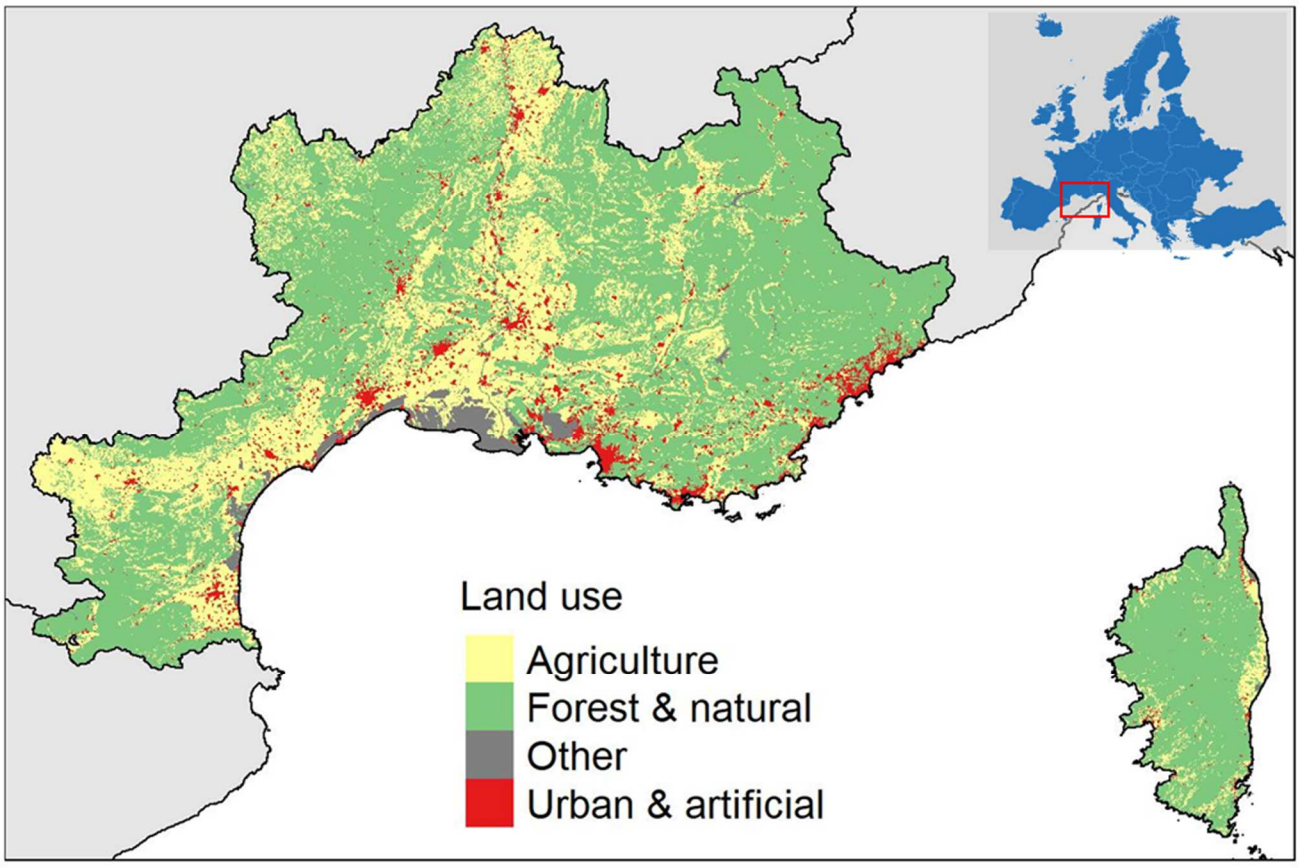
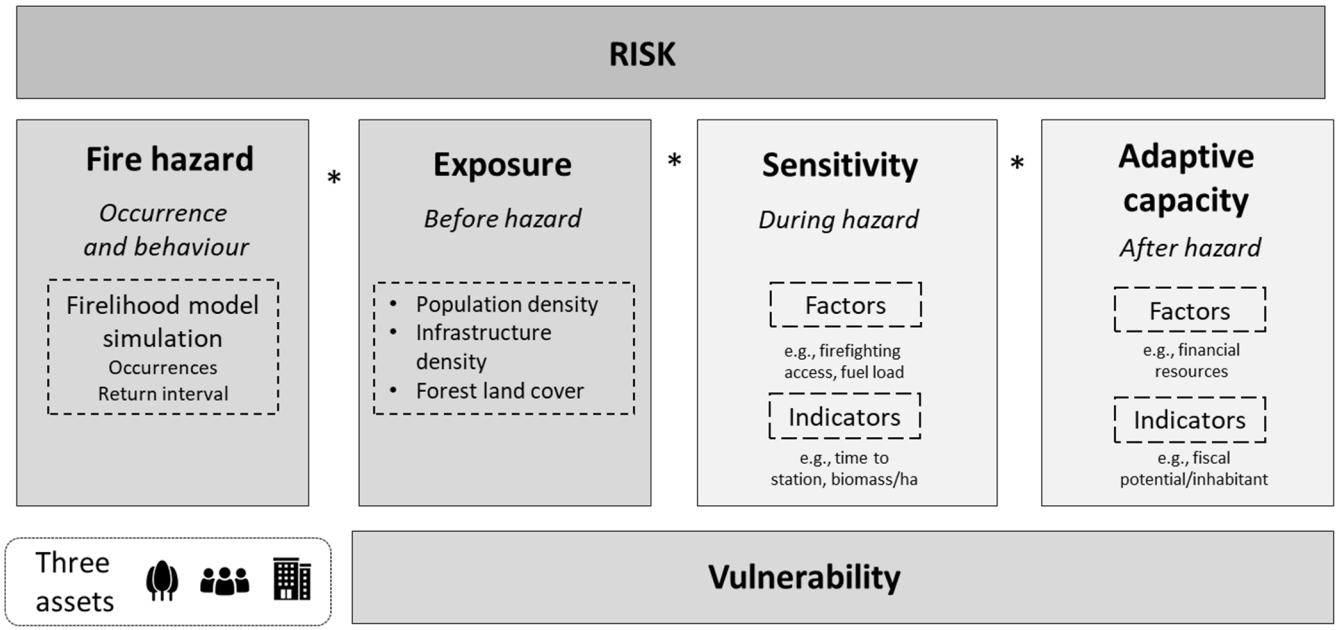


Figure 1 – Location and land-use in the study area. "Other" land-uses include wetlands, water bodies and open spaces with little to no vegetation in the Corine Landcover Classification.

2.1.2. Conceptual framework on vulnerability

We conceptualize fire risk and vulnerability following frameworks developed by Lecina-Diaz et al. (2021) and Oliveira et al. (2018) for forest disturbances and forest fires respectively (Figure 2). Risk and vulnerability are defined with regards to *exposed assets* (or values-at-risk) that contribute to wellbeing in societies and may be damaged. *Exposure* is defined as the presence of valuable assets before the hazard occurs. *Sensitivity* refers to the asset's susceptibility to sustain damage during or immediately after the hazard's occurrence. *Adaptive capacity* (AC) is defined as the asset's ability to cope with damage that has been sustained and to recover after the hazard has occurred. Both sensitivity and AC depend on *intrinsic factors* (e.g. bark thickness, housing density) and *extrinsic factors* (e.g. silvicultural practices, financial resources). *Vulnerability* then refers to the assets' propensity to be adversely affected by the hazard, to cope and recover from it, and is the resultant of exposure, sensitivity and AC. In turn, *risk* is defined as the potential for negative consequences for an asset due to a hazard that may occur. It is the resultant of vulnerability and *hazard*, which refers to the physical fire phenomenon (occurrence and behavior).



193 ***Figure 2 - Risk and vulnerability framework used in the assessment, adapted from Oliveira***
194 ***et al. (2018) and Lecina-Diaz et al. (2021)***

195 ### 2.1.3. The Analytical hierarchy process

196 The Analytical Hierarchy Process (AHP, Saaty 2004) is a weighted additive MCDA method
197 based on problem decomposition and the subsequent aggregation of indicators, where
198 weights are attributed to individual indicators through a binary comparison process involving
199 expert judgment. Within a GIS environment, AHP is used to compute weights attributed to
200 layers of spatial data representing criteria and to produce synthetic maps (Malczewski &
201 Rinner, 2015).

202 For each indicator pair, relative preference is quantified by experts using the verbal scale by
203 Saaty (2004), to which an underlying 1-9 scale is associated[2]. Scores are then compiled in a
204 matrix of order *n*, where c_{jk} is the pairwise comparison score between factors *j* and *k*, and *n* is
205 the number of factors. The matrix is reciprocal and scores on the diagonal are equal to 1,
206 hence only $n * (n-1)/2$ comparisons are required. The relative importance of factors, or
207 weights, are given by the principal eigenvector of the matrix. These are computed as $w_k =$
208 $\sum_{k=1}^{n} c_{jk}^{*}/n$, where w_k are the computed weights for factors and c_{jk}^{*} are the matrix's
209 coefficients normalized over columns. A consistency ratio *CR* is computed to assess the level
210 of inconsistency in expert judgments. It is computed as $(\lambda_{max} - n)/(RI * (n-1))$, where
211 λ_{max} is the maximum eigenvalue of the matrix and *RI*, or random index, is the consistency
212 ratio of randomly generated matrices.

213 Aggregation of individual judgements is performed using the geometric mean as it preserves
214 the AHP axioms and matrices' properties and dampens the effects of extreme judgements.

[2] See Appendix A.

Given a large enough sample size, consistency of the aggregated matrix will tend towards 1, e.g. a 20 sample size is enough to reach 90% consistency for matrices of order 3 or more (Aull-Hyde et al., 2006; Bernasconi et al., 2013). Once weights are computed, a score for the objective higher up in the hierarchy (e.g. sensitivity of population) is calculated using weighted summation. Within the GIS environment, scores are calculated by combining weights with layers of spatial data for each indicator and location.

2.2. Application to fire vulnerability

2.2.1. Problem decomposition and choice of indicators

We adopt a hierarchy with three levels (Fig. 2). The first level comprises the two concepts assessed in the study: risk and vulnerability. The second level corresponds to subcomponents of the two concepts: hazard, exposure, sensitivity and adaptive capacity (AC). The third level refers to factors, i.e. reasons why an asset may be more or less vulnerable, and indicators, i.e. variables developed to quantify factors. Factors are therefore defined by a rationale (e.g. a place that is easily accessible to firefighters is less sensitive), while indicators are measurable and characterized by values (e.g. distance to the nearest fire station).

Hazard

Hazard refers to the likelihood of fires, including both occurrence probability and fire spread. At the regional scale, fire activity can be represented through emergent propertie, in particular burned areas. Hazard was quantified using simulations with Firelihood, a probabilistic fire activity model developed for Southern France and estimated in a Bayesian framework using the Fire Weather Index (Van Wagner, 1987) and forest area as predictors (Pimont et al., 2021). It represents fire activity as a Poisson point process and comprises two sub-models: one determining fire occurrence probability and one determining fire sizes for

238 fires larger than 1ha. The sequential use of the two enables the simulation of fire occurrences
239 and their subsequent spread, yielding a spatialized account of burned areas at the 8km pixel
240 and daily scales. We used the mean annual burned area computed from a set of 1000
241 stochastic replications as the indicator for fire hazard.

242 ***Vulnerability***

243 For exposure, one indicator was developed per asset: population density, land-use share of
244 forest ecosystems and infrastructural building density. For sensitivity and adaptive capacity,
245 the choice of factors and indicators was based on a review of the specialized literature.
246 Factors also needed to be relevant at the regional study scale and quantifiable at the 1km
247 pixel scale sing publicly available data. For this reason, factors such as building materials
248 were discarded while date of construction was preferred. We give a brief overview of the
249 reasoning behind such choices below (Table 1).

250 Three factors for population's sensitivity relate to ease of evacuation (Table 1): the presence
251 of collective housing and the presence of large or single-adult households, who are harder to
252 evacuate, and the presence of young and elderly people who may require assistance.
253 Unemployment level and household wealth were selected as they relate to access to
254 resources, highlighted, contributing to social vulnerability (Emrich et al., n.d.; Geomatics et
255 al., 2017; Nguyen et al., 2017). Geographic isolation contributes to increased sensitivity by
256 making evacuation more complex while also relating to lower access to resources (services).
257 Several of these also relate to ease of defending the property (e.g. harder with children, easier
258 when better equipped). Factors for population's AC relate to their ability to recover over the
259 medium term by having access to various forms of community-level support (e.g. financial,
260 social), information and health services. Recent experience of fire events is also considered to
261 contribute to better AC (Edgeley et al., n.d.; Paveglio et al., 2009).

Three factors for ecosystem sensitivity were based on firefighting services' capacity to monitor, access and defend the area. Two other factors were based on forests' intrinsic characteristics: fuel load, known to favor fire activity and intensity, and the intrinsic sensitivity of tree species. Three factors for AC were based on extrinsic features, two of which are shared with other assets (risk experience and city-level financial resources). Besides, we considered public forests to have better AC given their active monitoring and management by authorities, while private forest owners, especially in fragmented and remote areas, can be absentee (Lindner et al., 2010). Two factors account for ecological features: tree species' intrinsic AC and fire return interval, estimated from the Firelihood model (Pimont et al., 2021).

Factors for the sensitivity of infrastructures relate to the buildings' capacity to withstand damage (recent buildings are designed for better evacuation and resistance), to ease of access for firefighting services as well as to the presence of sensitive industrial compounds nearby that may aggravate potential damages. We also took into account the presence of wildland-urban interface, i.e. intermingled vegetation and urban land-uses, which is known to favor fire activity (Ganteaume & Jappiot, 2013; Radeloff et al., 2018). Factors for the AC of infrastructures largely relate to the coping capacity of their owners, occupants and of local authorities, and we use the same factors as for populations, except access to health services and social isolation, which are specific to people.

2.2.2. Computation and aggregation

All indicators were computed on GIS software QGIS (https://qgis.org) at the scale of 1km wide pixels. They were computed from publicly available sociodemographic, economic and land-use data from the Geographic National Institute (IGN), the National Institute for Statistics and Economic Analysis (INSEE), the French Observatory of Territories, and the

French Forest Agency (ONF) databases. Most were natively at the pixel size (no treatment) or at larger pixel and municipality scales and were downscaled to 1km pixel size. Cover shares for land-use data were computed using QGIS's overlay and buffer tools, and distances using the nearest neighbor tool. Fuel load was directly taken from the raster map by the EU Joint Research Center (Avitabile & Camia, 2018) and computed with the raster calculator. Intrinsic sensitivity had been previously assessed for dominant tree species of the national high-resolution forest cover database (BD Forêt) by experts of the French Forest Agency (ONF) and was kindly provided as a raster map for this study.

Component	Factor	Indicator	Weight	CR
Asset 1 - Population				
Exposure	Resident population	Population density (people/km²) *	1	-
Sensitivity	Geographic isolation	Average access time to intermediate-level-services (min) •	0,42	1.40%
	Age	People under 10 and over 65 years old (%) *	0,18	
	Housing type	Households in collective housing (%) *	0,14	
	Household composition	Households with only one adult or 3 or more children (%) *	0,12	
	Unemployment	Unemployed people (%) *	0,07	
	Household wealth	Winsorised household wealth (€) *	0,06	
AC	Risk experience	Burned area over the past 10 years (ha, www.promethee.com)	0,30	1,90%
	Financial resources	Fiscal potential per inhabitant at city level (€) •	0,19	
	Access to health	Localised potential accessibility to doctors•	0,12	
	Social isolation	People either unemployed or over 75 years old (%) •	0,14	
	Geographic isolation	Average access time to intermediate-level-services (min) •	0,14	
	Education level	People with higher education (%) •	0,11	
Asset 2 - Ecosystems				
Exposure	Presence of forests	Pixel area covered by forests (%) †	1	-
Sensitivity	Instrinsic sensitivity	Pixel area with sensitivity classes 4-5 (%, ONF)	0,29	1.90%
	Fuel load	Biomass per hectare (Mt/ha, JRC Avitabile et al. 2020)	0,23	
	Surveillance	Pixel area covered by surveillance and patrol zones (%, ONF)	0,22	
	Denfendability	Pixel area within 200m of roads and water points (%, ONF)	0,19	
	Firefighting access	Distance to nearest station (km) †	0,07	
AC	Intrinsic adaptive capacity	Pixel area with adaptive capacity classes 4-5 (%, ONF)	0,43	0,80%
	Fire return interval	Fire return interval (years, firelihood model simulations Pimont et al. 2021)	0,29	
	Risk experience	Burned area over the past 10 years (ha, www.promethee.com)	0,10	
	Financial resources	Fiscal potential per inhabitant at city level (€) •	0.10	

	Property regime	Pixel area of publicly owned forests (%, www.data.gouv.fr)	0,07	
Asset 3 - Infrastructure				
Exposure	Presence of buildings	Pixel area covered by commercial, residential and industrial buildings (%) †	1	-
Sensitivity	Wildland-Urban Interface (WUI)	Pixel area within 200m of both forest and urban land-uses (%, CORINE Landcover)	0.50	0.50%
	Building use	Distance to closest Classified Installation of Environmental Protection ICPE (km, www.data.gouv.fr)	0,24	
	Firefighting access	Distance to nearest station (km) †	0,18	
	Age of building	Number of buildings built before 1990 (%)*	0,08	
AC	Financial resources	Fiscal potential per inhabitant at city level (€) •	0,54	2,10%
	Risk experience	Burned area over the past 10 years (ha, www.promethee.com)	0,23	
	Education level	People with higher education (%) •	0,12	
	Geographic isolation	Average access time to intermediate-level-services (min) •	0,11	
Fire Hazard				
Hazard	Mean annual burned area	Mean annual burned area (ha) over 1000 simulations with the fire model Firelihood (Pimont et al. 2021).	1	-

294

295 ***Table 1 – List of factors and indicators for risk and vulnerability components: exposure,***
296 ***sensitivity and adaptive capacity (AC). Indicators are presented by order of decreasing***
297 ***importance following the Analytical Hierarchy Process. Data was retrieved from publicly***
298 ***available databases from INSEE (*), Observatoire des Territoires (°) and IGN (+), unless***
299 ***indicated otherwise.***

300 Indicators use different units and scales and were standardized before summation using the
301 score range procedure, yielding unitless values between 0 and 1 (Malczewski, 1999). Most
302 were right-skewed and were winsorized between their 95% and 100% quantiles prior to
303 standardization. This procedure ensures that post-aggregation results use a large range of
304 values, facilitating map interpretation by experts in workshops. It amounts to assuming that
305 the contribution of any given indicator to vulnerability becomes marginal for very high
306 values at the right end of the distribution (e.g., that an extremely wealthy community is as
307 vulnerable as a very wealthy community).

308 After the weighted summation of indicators, vulnerability was obtained by multiplication of
309 exposure, sensitivity and AC. Risk was computed by multiplying vulnerability with hazard.
310 All metrics computed were represented on maps using the Jenks natural breaks algorithm, a
311 clustering procedure which minimizes variance within classes and maximizes variance across
312 classes and is particularly well suited for representing data on choropleth maps.

313 ### 2.3. Expert selection, interviews and discussion workshop

314 Given, the objective of bridging the gap between science and field knowledge, experts were
315 primarily drawn from local agencies working with wildfires and selected based on experience
316 rather than academic background. We interviewed policy officers from the French agencies
317 of the environment and agriculture and members of firefighting forces. We also included
318 scientists conducting research on fire risk. 48 experts were contacted and 22 agreed to be

interviewed (Table 2). Interviews lasted between 1 and 2 hours each and, given restrictions due to the Covid pandemic, were conducted through videoconference. The project's general objective and methods were explained to interviewees, examples of pairwise comparisons were given for training, and factors were explained beforehand.

Institution	Contacted	Interviewed
French forest agency	6	4
Firefighting and prevention	9	5
Public agencies	22	9
Scientists	7	3
Private sector	4	1
Total	48	22

Table 2 – Breakdown of experts interviewed.

The last step of the process was the organization of a workshop with the same pool of experts. As mapping constitutes a modified representation of a spatial phenomenon (Malczewski & Rinner, 2015; Palsky, 2013), resulting conclusions must be discussed by stakeholders to ensure results appropriation and to contribute to local risk management. this constitutes an *ex-post* validation of result where maps are considered as visualization and dialogue tools. We specifically discussed the geographical implications of results through a three steps process: (1) redefining key concepts, (2) collective appropriation of results, issue identification and (3) discussion on key messages and limitations for decision makers and practitioners. Inspired by participatory research approaches in a pandemic context (Hall et al.,

2021), workshops were conducted using the online collaborative workspace MURAL (https://www.mural.co/). This allowed participants to freely navigate through the maps and tables in an interactive manner while discussing results and writing down personal observations.

3. Results

3.1. AHP results and aggregation

AHP results show that experts considered geographic isolation to be the main factor contributing to population's sensitivity (42% weight) while all three factors related to ease of evacuation had intermediate weights (12-18%, total 43%). Access to resources (wealth, unemployment) was judged not to be as important (6-7%). Results for AC display moderate contributions for all factors (11-30%), among which past experience with fire hazard was judged most important (30%), followed by financial resources (19%). Experts judged that ecosystem sensitivity was primarily dependent on intrinsic forest characteristics - intrinsic tree sensitivity (29%) and fuel load (23%). As for factors related to firefighting, surveillance (22%) and defendability (19%) were deemed more important than ease of access (7%). AC was judged to be mainly determined by ecological factors, namely tree species' intrinsic adaptive capacity (43%) and fire return interval (29%), while socio-economic factors were deemed less relevant (7-10%). All matrices showed high levels of consistency with consistency indices below 5%. For infrastructures, sensitivity was judged by experts to be overwhelmingly dominated by the presence of WUI (50%). The presence of firefighting access, a mitigating factor, and that of sensitive industrial sites nearby, an aggravating factor, were also deemed important (18% and 24% respectively), while building age, an intrinsic characteristic, was not (8%). AC was judged to be mostly determined by financial resources (54%) and experience of risk (23%).

3.2. Vulnerability maps for individual assets

3.2.1. Population

Densely populated urban areas are mostly located along the coast and in the Rhône valley (Fig. 3a). In mountainous areas (e.g. the Alps), the most exposed areas are concentrated in valleys, while exposure is more homogeneous in the plains. Sparsely populated areas such as Corsica are little exposed. Sensitivity (Fig 4) is higher in the North-East and North-West of the study area as well as in Corsica. This corresponds to the most remote locations (largely mountainous), which are characterized by a high share of people with reduced mobility (esp. elderly) and the most distant to public and rescue services. Given the relatively low importance given to access to financial resources by experts, the South-West area, despite high unemployment and low wealth, appears as moderately sensitive. On the other hand, local clusters of intermediate sensitivity appear in well-connected urban centers (e.g. Narbonne, Marseille) owing in part to the high proportion of shared housing and large households[3].

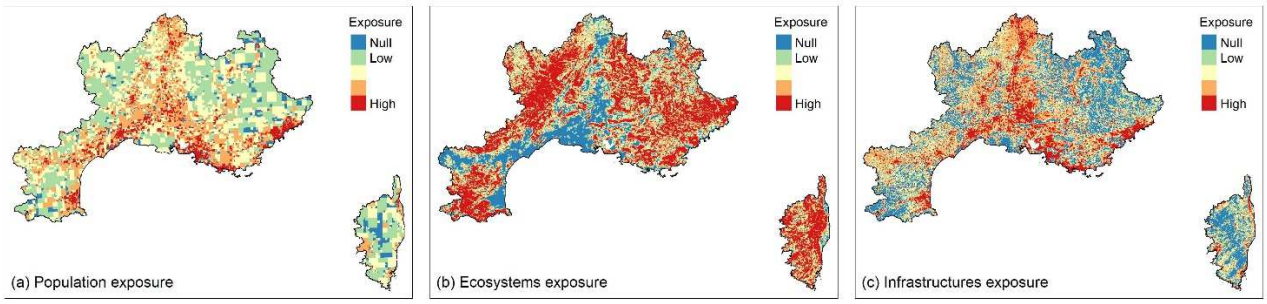
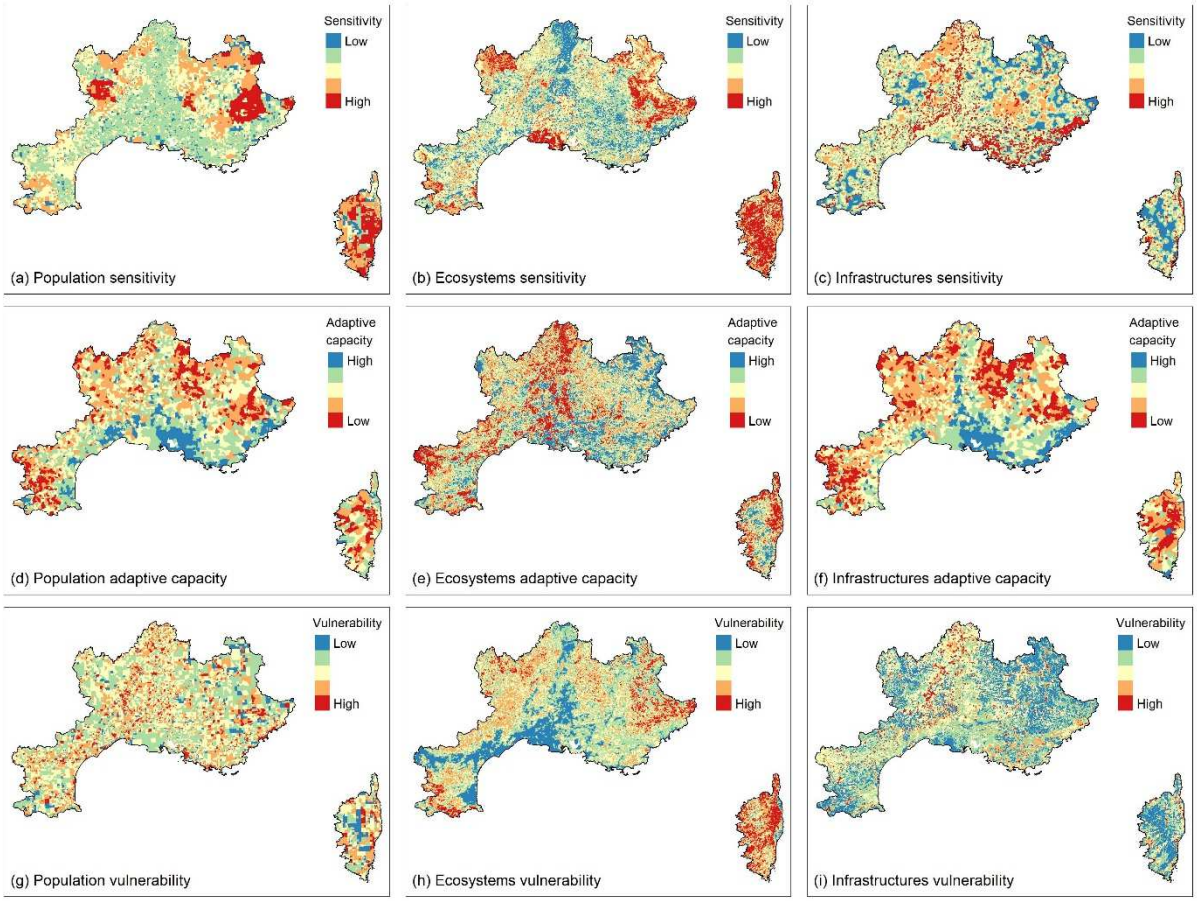


Figure 3 - Maps of exposure for the three assets studied: population (a), ecosystems (b) and infrastructures (c). Red areas indicate strong contribution to vulnerability (i.e. high exposure, cf. Table 1).

[3] Maps for individual indicators' contribution of vulnerability components are available as an Electronic Supplementary Material on the journal's website.

AC mostly follows the spatial patterns of risk experience and financial resources, with high values (i.e. low contribution to vulnerability, blue-green on maps) on cities on the eastern coast where both indicators display high values. Urban areas along the Rhône valley and the south-western coast display slightly lower AC due to only financial potential or risk experience being very strong respectively. Large urban areas (i.e. Marseille, Montpellier) display among the highest values for AC and are also characterized by high levels of education and access to health services. Lowest AC is found in the mountainous hinterland, where not only risk experience and financial resources are low, but also access to services, while social isolation is stronger. Despite low levels of social isolation and relatively high financial resources, the Northern Rhône and Alpine valleys display intermediate AC owing to the low importance given by experts to the factor, their low levels of risk experience and, for the latter, lower access to services.

Owing to low exposure despite high sensitivity and low AC, vulnerability is low or null in the hinterland and mountainous areas. On the contrary, urban areas along the Rhône valley and the coast, while strongly exposed, show limited vulnerability due to low sensitivity and higher AC. Areas where populations are most vulnerable are located at the margin between large urban centers and the hinterland, around secondary cities and along the Alps valleys, where sensitivity is high, AC low and exposure at an intermediate level. Figure 5 displays differences in vulnerability when using AHP weights instead of equal weights. Using expert knowledge increases vulnerability in remote areas (strong isolation), and decreases it along major population axes, to a limited extent.



396

397 ***Figure 4 - Maps of sensitivity, adaptive capacity and vulnerability for the three assets***
398 ***studied: population (a, d and g), ecosystems (b, e and h) and infrastructures (c, f and i).***
399 ***Red areas indicate strong contribution to vulnerability (i.e. high sensitivity, low AC).***
400 ***Larger maps with higher resolution can be found in electronic supplementary materials.***

401 ### *3.2.2. Ecosystems*

402 Ecosystem exposure matches the "forest and natural" type of land-use and therefore is
403 highest in hilly and mountainous areas located further away from urban centers, agricultural
404 land and vineyards.

405 Sensitivity is highest in areas that are both less surveilled and have high fuel loads, e.g.
406 remote mountainous locations to the North-East and North-West, but also in surveilled
407 locations with high fuel load and a strong presence of intrinsically sensitive forests, e.g., the

Ardèche, Cévennes and Vercors areas. Many areas with high intrinsic sensitivity, especially along the coast and its immediate hinterland, display moderate sensitivity values owing to lower levels of fuel load, surveillance of the area and proximity to fire stations. The moderating influence of defensible areas can be witnessed locally, e.g. in Corsica where their presence results in sensitivity values one or two classes lower than in similar, non-defensible locations. AC follows the general patterns of forests' intrinsic AC, even though some of these display low to moderate levels of AC owing to the influence of fire return interval, e.g. along the South-Western coast. Risk experience and financial resources intervene as aggravating or moderating factors locally.

Vulnerability is determined for low values by exposure, i.e., areas with little to no forests, including locations with low AC or high sensitivity (e.g. Rhône Valley). The highest values for vulnerability are found in elevated mountainous areas at the margins of the study area, where both exposure and sensitivity are strong while AC displays low to intermediate values. Hilly to mountainous areas along the Rhône river and along the South-Eastern coast display intermediate to high vulnerability due to a conjunction of AC and sensitivity. Within these areas, vulnerability hotspots (e.g. within the Ardèche and Cévennes) are driven by locally high values for sensitivity rather than low AC. The use of AHP weights leads to null or moderate decreases in assessed vulnerability in many locations, and moderate to strong increases in remaining locations, owing to complex interplays between indicators.

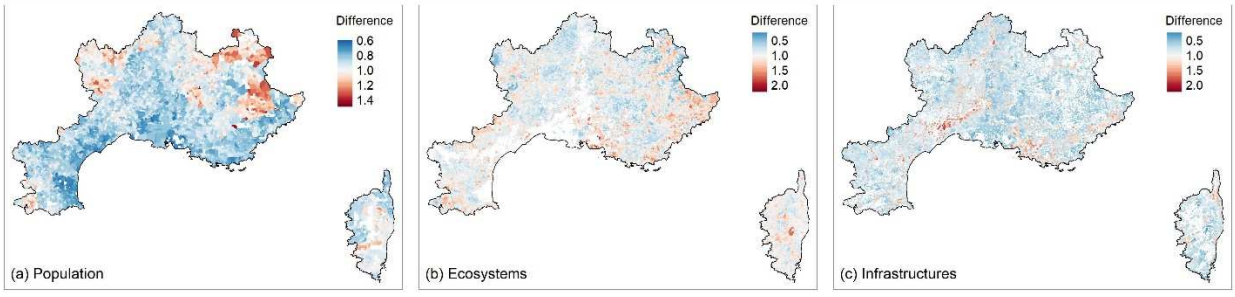


Figure 5 – Difference in vulnerability when using AHP weights compared to using equal weights for indicators. A value above a 1 means a higher vulnerability when using AHP weights (e.g. 1.5 means +50%).

3.2.3. Infrastructures

Exposure of infrastructures displays spatial patterns similar to that of population, i.e. high values along urban axes and a large share of pixels with null to very low values. However, it appears less concentrated as infrastructures may be encountered in rural areas despite low population rates for historical reasons (de-industrialization, rural exodus).

Sensitivity is highest along the Rhône valley and around the cities of Marseille and Nice, where it is driven by the presence of significant WUI, especially for the latter. Other locations with high sensitivity correspond to those close to industrial sites and located further away from fire stations (e.g. Ardèche, South-Western Alps, Western Languedoc). These locations also display a very high share of older buildings. Low sensitivity values are found locally in relatively isolated locations across the study area, where exposure is low.

AC is highest in southern Rhône Valley and along the South-Eastern coast owing to a conjunction of abundant financial resources and, to a lesser degree, high risk experience and education (Montpellier, Marseille). Low to intermediate AC values are found in rural areas with low financial resources. In such locations, high risk experience can be a moderating factor (e.g. Cévennes, Corsica) and low access to services an aggravating factor (e.g. southern Alps). AHP weights contribute to increasing assessed vulnerability in two types of locations: first around large urban centers of the South-Eastern coast (high WUI), second around sensitive industrial sites to the West of the Rhône valley, where financial resources are also lower while WUI is high.

3.3. Hazard and risk

Figure 6 (a) displays average burned areas from 1000 simulations with the Firelihood model as a metric for hazard. We observe hazard to be highest to the South of the study area, especially along the coast, in the Southern Rhône valley, the Cevennes area and Corsica, while mountainous locations are much less concerned.

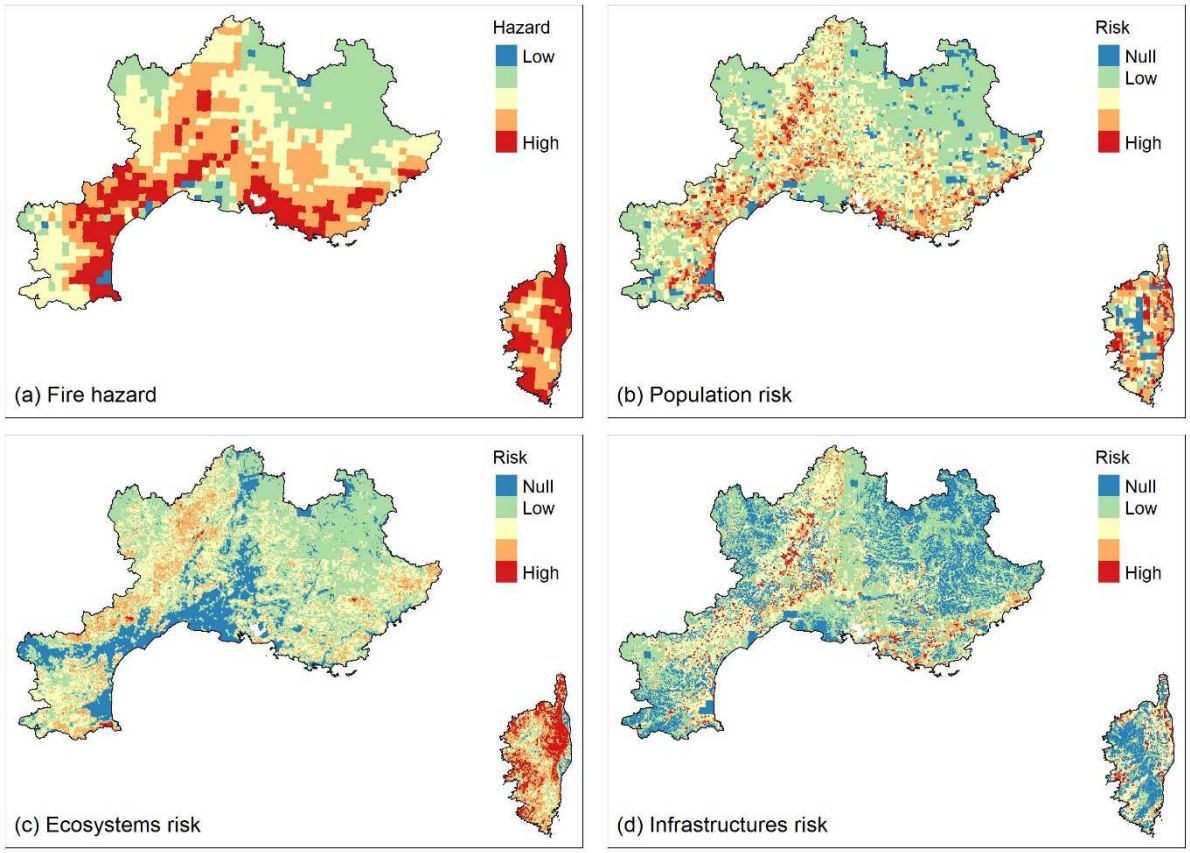


Figure 6 - Maps for (a) hazard (Firelihood model simulations), (b) population risk, (c) ecosystems risk and (d) infrastructures risk. Red areas indicate high risk.

Risk maps show that once hazard is considered, some areas display low risk levels despite being vulnerable owing to lower hazard levels. For ecosystems, this is the case of the mountainous Alps (north-east) and Pyrenees areas (south-west), and also in Aubrac (north-west, mostly due to sensitivity). While these locations tend not to be highly vulnerable for

463 population and infrastructures due to their relative isolation, this moderating influence of
464 hazard is striking for ecosystems. In the end, ecosystems risk is highest in the Ardèche and
465 especially Cevennes and South-Eastern coastal areas. Corsica displays high levels of both
466 vulnerability and risk for ecosystems.

467 Similarly, due to low hazard levels, the center of urban areas (e.g., the cities of Montpellier
468 and Perpignan) have low levels of risk. While vulnerability tends to already be null or low for
469 such locations for ecosystems (no exposure) and infrastructures (no industrial site), the
470 moderating influence on risk is higher for populations.

471

4. Discussion

4.1. Spatial patterns of vulnerability and risk and policy response

Our study provides a basis for discussing with stakeholders how to identify vulnerable regions, causative factors, and prioritize prevention and suppression measures. Besides, the separate consideration of population, ecosystems and infrastructures helps planning differentiated responses according to the importance given to each asset. A map-based approach may also improve how choices are made spatially and help choosing appropriate measures in different locations.

For example, while locations with highest vulnerability levels correspond to those already known to public authorities (e.g., coastline, Languedoc and Cevennes), our work suggests the existence of secondary vulnerability hotspots, often very localized, specific to one asset and caused by a limited number of factors. Besides, some areas also call for particular attention, such as Corsica, which appears to be very exposed but with a low adaptive capacity despite populations' strong experience of risk. This is a first step in identifying where more detailed assessments should be carried out with local services, or where specific policies (e.g. risk prevention) may be the preferred course of action

Results also showed that risk was moderate-to-low in some locations with high vulnerability but low hazard values. Given that fire regimes are expected to aggravate and reach new areas (Pimont et al., 2022, Dupuy et al., 2020), the assessment of exposure, sensitivity, adaptive capacity and their underlying factors in such locations is needed. Areas without past experience will become exposed, which represents a challenge for the implementation of public policies adapted not only to the characteristics of the territory but also to those of the populations.

4.2. Stakeholder participation and feedback

Our work also highlights the potential advantages of using expert knowledge in vulnerability studies applied to wildfire risk. Indeed, using the AHP aggregation also enabled us to get a quantified understanding of how sociodemographic and economic factors (e.g., risk experience, isolation), often overlooked in integrated assessments but known to be important in the specialized literature, compare to commonly found land-use, ecological and physical factors.

The method itself was highly appreciated by stakeholders, who emphasized the importance of including field-experts in applied research. Feedback stressed benefits both from a scientific point of view, as well as in terms of the actors' ability to reflect on their professional activity: "*This is very beneficial because it gives us an opportunity to think and pushes us out of our comfort zone. It makes us consider other types of indicators and wonder: what are the major factors to consider? It's interesting because it is questioning*[4]". MCDA became part of the knowledge creation process and gave birth to discussions on e.g. data availability and missing indicators to develop, several participants insisting on the potential value of long-term collaboration. Future endeavors could raise the question of integrating collective knowledge and local populations as well, probably at finer geographical scales (e.g. landscape level).

Regarding future evolution as well, experts underline the importance of the participatory approach for the exploration of subjects with little hindsight: "*Being associated is very important because there is little collaboration between operational services and research. This work brings technical and scientific evidence in a context of change where we do not always have straightforward answers*[5]". Our work contributes to it by helping identify

[4] Stakeholder's feedback from the workshop
[5] Stakeholder's feedback from the workshop

517 vulnerable areas even in the absence of hazard, and by inducing discussions on underlying
518 factors.

519 ### *4.3. Limitations and further research*

520 South-eastern France is heavily frequented by tourists during the fire season. Tourists are
521 particularly sensitive owing to their lower experience, harder access to information and
522 harder evacuation in heavily frequented locations. They were omitted owing to the difficulty
523 of deriving a simple indicator and of aggregating it to resident population numbers. Besides,
524 people are mobile and many casualties happen while fleeing, but we conceptualized exposure
525 as "before fire", our exposure metric was based on the town of residency, and proxies for
526 evacuation were independent of the fire event (e.g. housing type). Further work should strive
527 to consider evacuation in a dynamic manner, which may be easier at a finer geographical
528 scale, to include tourists. The inclusion of indicators related to ease of defense of a property
529 could also be improved, given that some people prefer to stay rather than flee (McLennan et
530 al., 2012).

531 While the conceptual framework derived from previous studies proved flexible for a
532 hierarchical MCDA approach, boundaries between the several components of risk and
533 vulnerability remain unclear. This is the case e.g. for factors such as defendability,
534 firefighting access or landscape structure, which were considered to contribute to sensitivity
535 but may also affect how fires ignite and behave, i.e., hazard. Setting boundaries remains
536 necessary for practical purposes, but choices made must be remembered when analyzing and
537 communicating about results. Besides, we followed several authors and computed
538 vulnerability as a product of its sub-components, with the rationale that when an asset is e.g.
539 not present, or does not sustain damage, it is not vulnerable. Others have adopted an additive

approach to vulnerability (Oliveira et al., 2018). This seems to remain a debated question in the conceptual literature.

While the participative and exploratory objectives of our work were attained, its non-predictive nature hinders further applied uses. While vulnerability maps provide powerful discussion tools, confidence in underlying values remains limited. Discussion-based subjective validation (e.g. interviews, workshops) may be enough for our purposes, but a more objective validation based on comparisons with external datasets would be necessary for prediction purposes. Besides, even though the AHP's consistency ratios were small enough, results remain contingent on experts interviewed, the choice of which can induce bias, and the small sample size hinders further analysis. To our knowledge, published vulnerability assessments do not exist for our study area, only evaluations of hazard and exposure, the former of which we already included. Besides, comparisons between model-based vulnerability values and composite indicators would be methodologically challenging owing to their distinct natures.

A subsequent and more ambitious effort would include a larger amount of field experts throughout the research project, from indicator selection to ex-post discussion. They could be drawn from several Mediterranean regions and professional backgrounds, enabling cross-comparisons across locations, expert profiles, and validation of indicators weights by comparing similar but independent expert groups. Methods such as fuzzy set theory could be used to decrease uncertainty arising from experts' interpretation of the AHP's linguistic elements, increasing confidence in indicator weights (Donevska et al., 2012). Other world regions subject to wildfire risk could also be studied (e.g. Australia, South America). Quantitative fire hazard evaluations are already performed there: applications of a participative methods would widen the scope of research conducted and provide insights on

564 how territorial vulnerability is perceived and defined based on different local experience,
565 environmental and socio-economic contexts.

566 Finally, our assessment was conducted at the regional scale and assets were considered as the
567 aggregation of many individual units. Some locations with a low number of sensitive and
568 non-resilient assets may appear as not vulnerable owing to a low exposure while, taken
569 individually, assets would be considered vulnerable (e.g. pixels with very few but isolated
570 and precarious elderly inhabitants). Our results should be taken as an assessment of
571 vulnerability at a collective level, helpful for decision makers to prioritize the allocation of
572 means, but not as an absolute evaluation. Other works should strive to develop a better
573 understanding of vulnerability dynamics at the individual level, e.g. individual people's
574 behaviors and factors of resilience.

575 ## 5. Conclusion

576 In this article, we developed a participative approach to assessing and mapping territorial
577 vulnerability to wildfires in South-Eastern France, distinguishing between several
578 components of vulnerability: exposure, sensitivity and adaptive capacity, and three assets
579 (populations, ecosystems and infrastructures), thus providing an encompassing overview of
580 the issue at stake. This hierarchical framework was coupled to the Analytical Hierarchy
581 Process to prioritize and spatially aggregate indicators taken from the specialized literature,
582 which proved a relevant and flexible way to integrate expert knowledge. Besides, model
583 simulation data was used to derive an indicator of fire hazard, demonstrating how classical
584 approaches to fire risk assessment could be used conjunctly to multi-criteria decision-making
585 approaches. Results helped identify vulnerable locations overall and for each asset
586 individually. We were also able to highlight major causative factors, and e.g. confirmed the
587 importance of the wildland-urban interface and of experience with past fires. Secondary

588 vulnerability hotspots were revealed, as well as vulnerable locations not yet confronted to
589 hazard but where prevention efforts may be needed with respect to climate change. The
590 participative approach also proved a successful manner of engaging stakeholder discussion.
591 Interviews with experts, even during the covid pandemic, enabled discussions to take place,
592 and an ex-post workshop was organized to put results into perspective and get feedback on
593 the process. Field experts were in particular pleased with the introduction of the adaptive
594 capacity concept and the inclusion of socio-economic and demographic data, which,
595 according to them, encouraged novel ways of thinking. Further research should include
596 several fire-prone regions and a larger number of experts, enabling comparisons across expert
597 groups, locations, and the external validation of indicator weights to allow for more applied
598 uses.

Credit author statement

Miguel Rivière: Conceptualization, Methodology, Formal analysis, Data curation, Writing – Original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition

Jonathan Lenglet: Conceptualization, Methodology, Writing – Original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition

Adrien Noirault: Formal analysis, Investigation, Data curation, Visualization

François Pimont : Methodology, Software, Writing – review & editing

Jean-Luc Dupuy : Methodology, Software, Writing – review & editing

Funding

This work is supported by a grant overseen by the French National Research Agency (ANR) as part of the "Investissements d'Avenir" program (ANR-11-LABX-0002-01, Lab of Excellence ARBRE).

References

Aull-Hyde, R., Erdogan, S., & Duke, J. M. (2006). An experiment on the consistency of aggregated comparison matrices in AHP. *European Journal of Operational Research*, *171*(1), 290–295. https://doi.org/10.1016/J.EJOR.2004.06.037

Avitabile, V., & Camia, A. (2018). An assessment of forest biomass maps in Europe using harmonized national statistics and inventory plots. *Forest Ecology and Management*, *409*, 489–498. https://doi.org/10.1016/j.foreco.2017.11.047

Bernasconi, M., Choirat, C., Operational, R. S.-E. J. of, & 2014, undefined. (2013). Empirical properties of group preference aggregation methods employed in AHP: Theory and evidence. *Elsevier*. https://doi.org/10.1016/j.ejor.2013.06.014

Bowman, D. M. J. S., Kolden, C. A., Abatzoglou, J. T., Johnston, F. H., van der Werf, G. R., & Flannigan, M. (2020). Vegetation fires in the Anthropocene. *Nature Reviews Earth & Environment*, *1*(10), 500–515. https://doi.org/10.1038/s43017-020-0085-3

Castel-Clavera, J., Pimont, F., Opitz, T., Ruffault, J., Rivière, M., & Dupuy, J.-L. (2022). Disentangling the factors of spatio-temporal patterns of wildfire activity in south-eastern France. *International Journal of Wildland Fire*, *32*(1), 15–28. https://doi.org/10.1071/wf22086

Comte, A., Pendleton, L. H., Bailly, D., & Quillérou, E. (2019). Conceptual advances on global scale assessments of vulnerability: Informing investments for coastal populations at risk of climate change. *Marine Policy*, *99*, 391–399. https://doi.org/10.1016/J.MARPOL.2018.10.038

Costafreda-Aumedes, S., Comas, C., & Vega-Garcia, C. (2017). Human-caused fire occurrence modelling in perspective: A review. In *International Journal of Wildland Fire* (Vol. 26, Issue 12, pp. 983–998). https://doi.org/10.1071/WF17026

Diaz, S., Settele, J., Brondizio, E., Ngo, H. T., Guèze, M., Agard Trinidad, J., Arneth, A., Balvanera, P., Brauman, K., Watson, R., Baste, I., Larigauderie, A., Leadley, P., Pascual, U., Baptiste, B., Demissew, S., Dziba, L., Erpul, G., Fazel, A., … Mooney, H. (2019). *Summary for policymakers of the global assessment report on biodiversity and ecosystem services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services-ADVANCE UNEDITED VERSION-Members of the management committee who provid.* https://www.ipbes.net/system/tdf/spm_global_unedited_advance.pdf?file=1&type=node&id=35245

Donevska, K. R., Gorsevski, P. V., Jovanovski, M., & Peševski, I. (2012). Regional non-hazardous landfill site

selection by integrating fuzzy logic, AHP and geographic information systems. *Environmental Earth Sciences*, *67*(1), 121–131. https://doi.org/10.1007/s12665-011-1485-y

Dupuy, J. luc, Fargeon, H., Martin-StPaul, N., Pimont, F., Ruffault, J., Guijarro, M., Hernando, C., Madrigal, J., & Fernandes, P. (2020). Climate change impact on future wildfire danger and activity in southern Europe: a review. In *Annals of Forest Science* (Vol. 77, Issue 2, p. 35). https://doi.org/10.1007/s13595-020-00933-5

Duzgun, H. S. B., Yucemen, M. S., Kalaycioglu, H. S., Celik, K., Kemec, S., Ertugay, K., & Deniz, A. (2011). An integrated earthquake vulnerability assessment framework for urban areas. *Natural Hazards*, *59*(2), 917–947. https://doi.org/10.1007/S11069-011-9808-6

Edgeley, C., reduction, T. P.-I. journal of disaster risk, & 2017, undefined. (n.d.). Community recovery and assistance following large wildfires: The case of the Carlton Complex Fire. *Elsevier*. Retrieved November 21, 2022, from https://www.sciencedirect.com/science/article/pii/S2212420917302418?casa_token=fk0LH_RcGRMAAAAA:TKTW_4Cy2DkFRGmm6txJzlHrM6k1u5h70KYOnkph2ICGhk6bqDBxQ4Y0RvN7tsWyAInst7E

Emrich, C., Weather, S. C.-, Climate, undefined, Society, and, & 2011, undefined. (n.d.). Social vulnerability to climate-sensitive hazards in the southern United States. *Journals.Ametsoc.Org*. https://doi.org/10.1175/2011WCAS1092.1

Fernandez, P., Mourato, S., & Moreira, M. (2015). Social vulnerability assessment of flood risk using GIS-based multicriteria decision analysis. A case study of Vila Nova de Gaia (Portugal). *Http://Www.Tandfonline.Com/Action/JournalInformation?Show=aimsScope&journalCode=tgnh20#.VsXodSCLRhE*, *7*(4), 1367–1389. https://doi.org/10.1080/19475705.2015.1052021

Finney, M. A. (2005). The challenge of quantitative risk analysis for wildland fire. *Forest Ecology and Management*, *211*(1–2), 97–108. https://doi.org/10.1016/j.foreco.2005.02.010

Füssel, H. M., & Klein, R. J. T. (2006). Climate change vulnerability assessments: An evolution of conceptual thinking. *Climatic Change*, *75*(3), 301–329. https://doi.org/10.1007/S10584-006-0329-3

Ganteaume, A., Camia, A., Jappiot, M., San-Miguel-Ayanz, J., Long-Fournel, M., & Lampin, C. (2013). A review of the main driving factors of forest fire ignition over Europe. In *Environmental Management* (Vol. 51, Issue 3, pp. 651–662). Springer. https://doi.org/10.1007/s00267-012-9961-z

Ganteaume, A., & Jappiot, M. (2013). What causes large fires in Southern France. *Forest Ecology and Management*, *294*, 76–85. https://doi.org/10.1016/j.foreco.2012.06.055

Geomatics, F. R.-, Risk, N. H. and, & 2017, undefined. (2017). Flood risk management centred on clusters of territorial vulnerability. *Taylor & Francis*, *8*(2), 525–543. https://doi.org/10.1080/19475705.2016.1250111

Ghajari, Y. E., Alesheikh, A. A., Modiri, M., Hosnavi, R., & Abbasi, M. (2017). Spatial modelling of urban physical vulnerability to explosion hazards using GIS and fuzzy MCDA. *Mdpi.Com*. https://doi.org/10.3390/su9071274

Ghorbanzadeh, O., Blaschke, T., Gholamnia, K., & Aryal, J. (2019). Forest Fire Susceptibility and Risk Mapping Using Social/Infrastructural Vulnerability and Environmental Variables. *Fire 2019, Vol. 2, Page 50*, *2*(3), 50. https://doi.org/10.3390/FIRE2030050

Gill, A. M., Stephens, S. L., & Cary, G. J. (2013). The worldwide "wildfire" problem. *Ecological Applications*, *23*(2), 438–454. https://doi.org/10.1890/10-2213.1

Hall, J., Gaved, M., & Sargent, J. (2021). Participatory Research Approaches in Times of Covid-19: A Narrative Literature Review. *International Journal of Qualitative Methods*, *20*. https://doi.org/10.1177/16094069211010087

IPCC. (2019). *IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security and Greenhouse gas fluxes in Terrestrial Ecosystems*. https://philpapers.org/rec/SHUCCA-2

IPCC. (2021). *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (V. Masson-Delmotte, P. Zhai, A. Pirani, S. Connors, S. Péan, S. Berger, N. Caud, Y. CHen, L. Goldfarb, M. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. Matthews, T. Maycock, T. Waterfield, O. Yeleki, R. Yu, & B. Zhou (eds.); Cambridge).

Johnston, L. M., Wang, X., Erni, S., Taylor, S. W., Mcfayden, C. B., Oliver, J. A., Stockdale, C., Christianson, A., Boulanger, Y., Gauthier, S., Arseneault, D., Wotton, B. M., Parisien, M.-A., & Flannigan, M. D. (2020). Wildland fire risk research in Canada. *Environmental Reviews*. https://doi.org/10.1139/er-2019-0046

Le Cozannet, G., Garcin, M., Bulteau, T., Mirgon, C., Yates, M. L., Méndez, M., Baills, A., Idier, D., & Oliveros, C. (2013). Climate of the Past Geoscientific Instrumentation Methods and Data Systems Solid Earth An AHP-derived method for mapping the physical vulnerability of coastal areas at regional scales. *Hazards Earth Syst. Sci*, *13*, 1209–1227. https://doi.org/10.5194/nhess-13-1209-2013

Lecina-Diaz, J., Martínez-Vilalta, J., Alvarez, A., Banqué, M., Birkmann, J., Feldmeyer, D., Vayreda, J., &

Retana, J. (2021). Characterizing forest vulnerability and risk to climate-change hazards. *Frontiers in Ecology and the Environment*, *19*(2), 126–133. https://doi.org/10.1002/FEE.2278

Lecina-Diaz, J., Martínez-Vilalta, J., Alvarez, A., Vayreda, J., & Retana, J. (2021). Assessing the Risk of Losing Forest Ecosystem Services Due to Wildfires. *Ecosystems*, *24*(7), 1687–1701. https://doi.org/10.1007/S10021-021-00611-1

Lindner, M., Maroschek, M., Netherer, S., Kremer, A., Barbati, A., Garcia-Gonzalo, J., Seidl, R., Delzon, S., Corona, P., Kolström, M., Lexer, M. J., & Marchetti, M. (2010). Climate change impacts, adaptive capacity, and vulnerability of European forest ecosystems. *Forest Ecology and Management*, *259*(4), 698–709. https://doi.org/10.1016/j.foreco.2009.09.023

Liu, J. C., Pereira, G., Uhl, S. A., Bravo, M. A., & Bell, M. L. (2015). A systematic review of the physical health impacts from non-occupational exposure to wildfire smoke. *Environmental Research*, *136*, 120–132. https://doi.org/10.1016/J.ENVRES.2014.10.015

Liu, X. peng, Zhang, J. quan, Tong, Z. jun, & Bao, Y. (2012). GIS-based multi-dimensional risk assessment of the grassland fire in northern China. *Natural Hazards*, *64*(1), 381–395. https://doi.org/10.1007/s11069-012-0244-z

Malczewski, J. (1999). *GIS and multicriteria decision analysis*. https://books.google.com/books?hl=en&lr=&id=ZqUsEAAAQBAJ&oi=fnd&pg=PA393&dq=malczewski+mcda+1999&ots=6H088cmAa0&sig=0bGVd-IlS2tszY42uabV7zp47XY

Malczewski, J. (2007). GIS-based multicriteria decision analysis: a survey of the literature. *Https://Doi.Org/10.1080/13658810600661508*, *20*(7), 703–726. https://doi.org/10.1080/13658810600661508

Malczewski, J., & Rinner, C. (2015). *Multicriteria Decision Analysis in Geographic Information Science*. https://doi.org/10.1007/978-3-540-74757-4

McLennan, J., Elliott, G., & Omodei, M. (2012). Householder decision-making under imminent wildfire threat: Stay and defend or leave? *International Journal of Wildland Fire*, *21*(7), 915–925. https://doi.org/10.1071/WF11061

Mendoza, G. A., & Martins, H. (2006). Multi-criteria decision analysis in natural resource management: A critical review of methods and new modelling paradigms. In *Forest Ecology and Management* (Vol. 230, Issues 1–3, pp. 1–22). Elsevier. https://doi.org/10.1016/j.foreco.2006.03.023

Metzger, M. J., Rounsevell, A., Acosta-Michlik, L., Leemans, R., & Schröter, D. (2006). *The vulnerability of*

ecosystem services to land use change. https://doi.org/10.1016/j.agee.2005.11.025

Nguyen, C. V., Horne, R., Fien, J., & Cheong, F. (2017). Assessment of social vulnerability to climate change at the local scale: development and application of a Social Vulnerability Index. *Climatic Change*, *143*(3–4), 355–370. https://doi.org/10.1007/S10584-017-2012-2

Nyimbili, P. H., Erden, T., & Karaman, H. (2018). Integration of GIS, AHP and TOPSIS for earthquake hazard analysis. *Natural Hazards*, *92*(3), 1523–1546. https://doi.org/10.1007/S11069-018-3262-7

O'Brien, J. J., Hiers, J. K., Varner, J. M., Hoffman, C. M., Dickinson, M. B., Michaletz, S. T., Loudermilk, E. L., & Butler, B. W. (2018). Advances in Mechanistic Approaches to Quantifying Biophysical Fire Effects. *Current Forestry Reports*, *4*(4), 161–177. https://doi.org/10.1007/S40725-018-0082-7

Oliveira, S., Félix, F., Nunes, A., Lourenço, L., Laneve, G., & Sebastián-López, A. (2018). Mapping wildfire vulnerability in Mediterranean Europe. Testing a stepwise approach for operational purposes. *Journal of Environmental Management*, *206*, 158–169. https://doi.org/10.1016/J.JENVMAN.2017.10.003

Oliveira, S., Gonçalves, A., Benali, A., Sá, A., Zêzere, J. L., & Pereira, J. M. (2020). Assessing Risk and Prioritizing Safety Interventions in Human Settlements Affected by Large Wildfires. *Forests 2020, Vol. 11, Page 859*, *11*(8), 859. https://doi.org/10.3390/F11080859

Oliveira, S., Rocha, J., & A Sá. (2021). Wildfire risk modeling. *Current Opinion in Environmental Science*. https://www.sciencedirect.com/science/article/pii/S2468584421000465?casa_token=vqt8Bc0csyMAAAA A:DzFyJMCdZRAGRv7geiR1kQgPMDolq2hSI3NeW0QKfeaW1v9lZOetdqTx8xoBkCEiFCJ63UgDlBE

Palsky, G. (2013). Cartographie participative, cartographie indisciplinée. *L'Information Géographique*, *Vol. 77*(4), 10–25. https://doi.org/10.3917/lig.774.0010

Paveglio, T., Brenkert-Smith, H., Fire, T. H.-… of W., & 2015, undefined. (2009). Understanding social impact from wildfires: Advancing means for assessment. *CSIRO Publishing*. https://doi.org/10.1071/WF14091

Pham, B. T., Luu, C., Dao, D. Van, Phong, T. Van, Nguyen, H. D., Le, H. Van, von Meding, J., & Prakash, I. (2021). Flood risk assessment using deep learning integrated with multi-criteria decision analysis. *Knowledge-Based Systems*, *219*, 106899. https://doi.org/10.1016/J.KNOSYS.2021.106899

Pimont, F., Fargeon, H., Opitz, T., Ruffault, J., Barbero, R., Riviere, M., & Dupuy, J.-L. (2021). Prediction of regional wildfire activity in the probabilistic Bayesian framework of Firelihood. *Ecological Applications*.

Pimont, F., Ruffault, J., Opitz, T., Fargeon, H., Castel-Clavera, J., Martin-StPaul, N., Rigolot, E., Barbero, R., & Dupuy, J.-L. (2022). Lengthening, expansion and intensification of future fire activities in South-Eastern France. In *Advances in Forest Fire Research 2022* (pp. 1198–1203). CSIRO PUBLISHING.

https://doi.org/10.14195/978-989-26-2298-9_181

Radeloff, V. C., Helmers, D. P., Anu Kramer, H., Mockrin, M. H., Alexandre, P. M., Bar-Massada, A., Butsic, V., Hawbaker, T. J., Martinuzzi, S., Syphard, A. D., & Stewart, S. I. (2018). Rapid growth of the US wildland-urban interface raises wildfire risk. *Proceedings of the National Academy of Sciences of the United States of America*, *115*(13), 3314–3319. https://doi.org/10.1073/PNAS.1718850115

Saaty, T. L. (2004). Decision making — the Analytic Hierarchy and Network Processes (AHP/ANP). *Journal of Systems Science and Systems Engineering*, *13*(1), 1–35. https://doi.org/10.1007/S11518-006-0151-5

Seidl, R., Thom, D., Kautz, M., Martin-Benito, D., Peltoniemi, M., Vacchiano, G., Wild, J., Ascoli, D., Petr, M., Honkaniemi, J., Lexer, M. J., Trotsiuk, V., Mairota, P., Svoboda, M., Fabrika, M., Nagel, T. A., & Reyer, C. P. O. (2017). Forest disturbances under climate change. In *Nature Climate Change* (Vol. 7, Issue 6, pp. 395–402). Nature Publishing Group. https://doi.org/10.1038/nclimate3303

Thompson, M. P., Scott, J., Helmbrecht, D., & Calkin, D. E. (2013). Integrated wildfire risk assessment: Framework development and application on the lewis and clark national forest in Montana, USA. *Integrated Environmental Assessment and Management*, *9*(2), 329–342. https://doi.org/10.1002/IEAM.1365

Thorne, J. H., Choe, H., Stine, P. A., Chambers, J. C., Holguin, A., Kerr, A. C., Schwartz, M. W., & Muir, J. (2010). Climate change vulnerability assessment of forests in the Southwest USA. *Climatic Change*, *148*, 387–402. https://doi.org/10.1007/s10584-017-2010-4

Vaillant, N. M., Kolden, C. A., & Smith, A. M. S. (2016). Assessing Landscape Vulnerability to Wildfire in the USA. In *Current Forestry Reports* (Vol. 2, Issue 3, pp. 201–213). Springer International Publishing. https://doi.org/10.1007/s40725-016-0040-1

Van Wagner, C. E. (1987). Development and structure of the Canadian forest fire weather index system. In *Forestry* (Vol. 51, Issue 3). https://doi.org/19927

Xi, D. D. Z., Taylor, S. W., Woolford, D. G., & Dean, C. B. (2019). Statistical Models of Key Components of Wildfire Risk. *Annual Review of Statistics and Its Application*, *6*(1), 197–222. https://doi.org/10.1146/annurev-statistics-031017-100450

Appendix A

Table A.1 – Scale used for pairwise comparison of factors (from Saaty 2004)

Score	Definition
1	Equal importance
3	Moderate importance of one over the other
5	Essential or strong importance
7	Very strong importance
9	Extreme importance
2, 4, 6, 8	Intermediate values between two adjacent judgements
Reciprocals	If factor i has one of the score assigned to it when compared with factor j, then j has the reciprocal score assigned to it when compared to factor i.

Appendix B

Figure B.1 – Annotated land-use maps of the study area

